

area

proof: $T = T_1 \cup T_2$, $\therefore T_1, T_2 \in \mathcal{M}$, $\therefore T \in \mathcal{M}$. 3. $a(\text{point } P) = 0$
 $a(T) = a(T_1) + a(T_2)$ (from 2)
 $= \frac{1}{2}b_1h + \frac{1}{2}b_2h$
 $= \frac{1}{2}bh$.



$\Rightarrow T \in \mathcal{M}$, $a(T) = \frac{1}{2}bh$

proof: $T \in \mathcal{M}$ (from 7.2)

$\therefore a(T \cup T_3) = a(T) + a(T_3)$ (from 2)
 $\therefore a(T) = a(T \cup T_3) - a(T_3)$

$$= \frac{1}{2}(b+b_3)h - \frac{1}{2}b_3h$$

$$= \frac{1}{2}bh.$$

8 P: $\Rightarrow P \in \mathcal{M}$, $a(P) = bh$

proof: $\therefore P = T_1 \cup T_2$, $T_1, T_2 \in \mathcal{M}$, $\therefore P \in \mathcal{M}$. $\therefore a(P) = a(T_1) + a(T_2)$ (from 2)
 $= 2a(T_2) = bh$

9. trapezoid T: $\Rightarrow T \in \mathcal{M}$, $a(T) = \frac{1}{2}(b_1+b_2)h$. $\therefore a(T) = \frac{1}{2}a(R_1) = \frac{1}{2}bh$



proof: $\therefore T = T_1 \cup T_2$, $T_1, T_2 \in \mathcal{M}$. 7.2 $a(T) = a(T_1) + a(T_2) + a(T_3)$ (from 2)
 $= \frac{1}{2}b_1h + b_2h + \frac{1}{2}b_3h = \frac{1}{2}(b_1+b_3)h$. (here $b_2 = b$)

4. $a(\text{line } L) = 0$
 $a(\bigcup_i R_i) = 0$, R_i point (use 2)

5. $a(\bigcup_i L_i) = 0$, L_i line (use 2)

step region $R = \bigcup_{i=1}^n R_i$,
 R_i rectangle, $R_i \neq R_j$ if $i \neq j$

$$\Rightarrow a(R) = \sum_{i=1}^n a(R_i) \quad (\text{use 2})$$

6. $T: \text{rhombus}$ $\Rightarrow T \in \mathcal{M}$, $a(T) = \frac{1}{2}bh$



proof: $\therefore T = R_1 \cap R_2$, R_1, R_2 rec
 $\therefore T \in \mathcal{M}$.

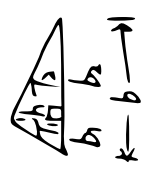
$\therefore T$ congruent T' , $\therefore a(T) = a(T')$

$$a(R) = a(T \cup T') = a(T) + a(T') \quad (\text{use 2})$$

$$= 2a(T)$$

$$\therefore a(T) = \frac{1}{2}a(R_1) = \frac{1}{2}bh$$

7. T triangle $\Rightarrow T \in \mathcal{M}$, $a(T) = \frac{1}{2}bh$



7.2 $\Rightarrow T \in \mathcal{M}$, $a(T) = \frac{1}{2}bh$

1. $S_i \in \mathcal{M}$, $i=1, 2, \dots, n$.

$$\Rightarrow \bigcup_i S_i \in \mathcal{M}, \quad \forall S_i \in \mathcal{M}, a(\bigcup_i S_i) \leq \sum a(S_i)$$

proof: $S_1 \cup S_2 \in \mathcal{M}$, $S_1 \cap S_2 \in \mathcal{M}$, $a(S_1 \cup S_2) \leq a(S_1) + a(S_2)$

if $S_1 \cap S_2 \in \mathcal{M}$, $S_1 \cap S_2 \in \mathcal{M}$, $a(S_1 \cup S_2) \leq a(S_1) + a(S_2)$

then, $S_1 \cup S_2 \cup S_3 \in \mathcal{M}$, $S_1 \cap S_2 \cap S_3 \in \mathcal{M}$,

$$a(S_1 \cup S_2 \cup S_3) \leq a(S_1 \cup S_2) + a(S_3)$$

$$\leq a(S_1) + a(S_2) + a(S_3) \quad \text{IV}$$

2. $a(S_i \cap S_j) = 0$, $\forall i \neq j$. here $S_i \in \mathcal{M}$, $i=1, 2, \dots, n$.

$$\Rightarrow a(\bigcup_i S_i) = \sum a(S_i)$$

proof: $a(S_1 \cup S_2) = a(S_1) + a(S_2)$

$$= a(S_1) + a(S_2)$$

$$\therefore a(S_1 \cup S_2) = a(S_1) + a(S_2)$$

then, $a(S_1 \cup S_2 \cup S_3) = a(S_1 \cup S_2) + a(S_3) - a(S_1 \cap S_2 \cup S_3)$

$$\therefore a(S_1 \cup S_2 \cup S_3) = a\left(\bigcup_{i=1}^3 (S_i \cap S_3)\right)$$

$$\leq \sum_{i=1}^3 a(S_i \cap S_3) = 0$$

$$\therefore a(S_1 \cup \dots \cup S_n) = a(S_1) + \dots + a(S_n) \quad \text{IV}$$